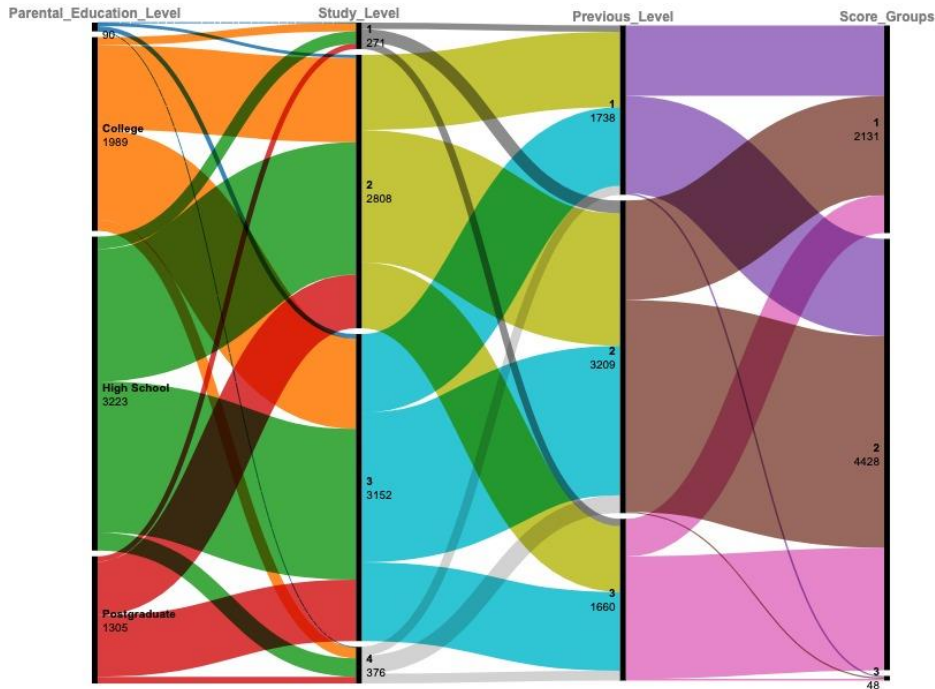


Data Visualization Pitch

Student Individual Assignment



The impact of different factors on student performances

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Research questions

Primary research question:

Which different factors impact students' performances?

Secondary research question:

- Can we reduce the matter of students performances only to how many hours do they spend studying?
- Is there an association between perceived teacher quality and student performance?
- Can we find a pattern in students' background that shows which students tend to perform better?

About Data

This dataset contains informations from 6607 students and includes 20 attributes related to their personal habits, previous academic performances and family background.

LICENSE:

Data from <https://www.kaggle.com/datasets/ayeshaseherr/student-performance> with CC BY-SA 4.0 license:

- -BY: you can share and adapt the work as long as you give appropriate credit to the author.
- -SA: you can share and adapt the work as long as you share the work you derive from it with the same license.

DATA QUALITY AND CRITICALITY:

The cleaning of this dataset didn't require any actions since there weren't any NA but while studying the variable "Teacher Quality" there were 78 empty records; while representing this variable those record were excluded.

The main criticality encountered during the acquisition of the data is that The Student Performance Factors dataset was obtained from Kaggle did not specify an original data collection methodology or specific population. This means that we can still understand some interesting insights about students performances but we can't locate them in any certain area.

Methodology

This Dataset was downloaded manually from Kaggle.

The Processing & Cleaning phase was carried out with RStudio using the libraries Tidyverse and dplyr.

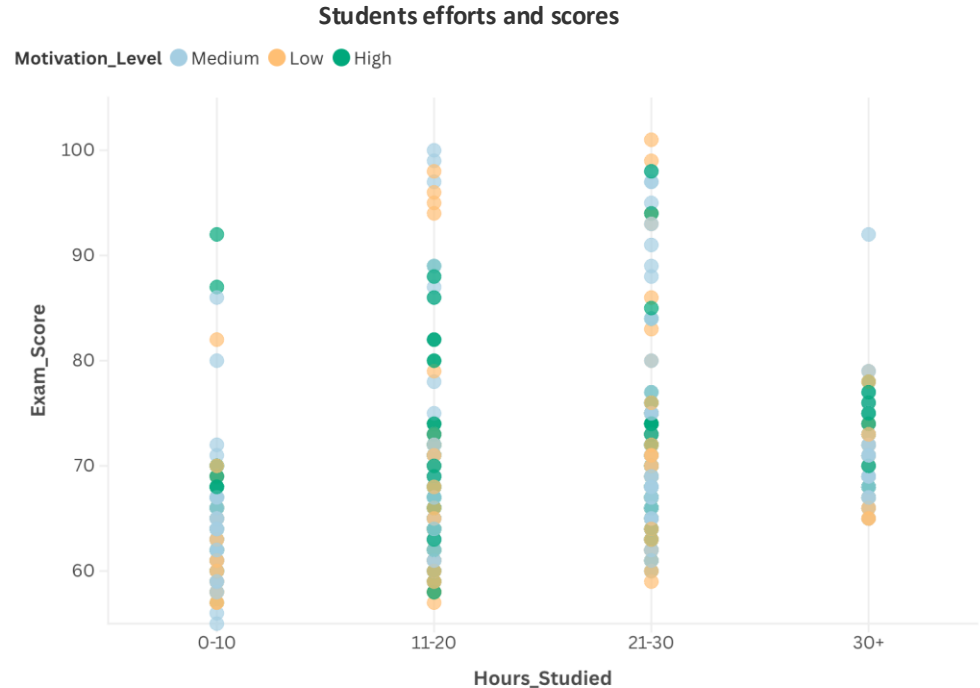
Data Transformation: To have a more meaningful representation, three numerical variables (Hours_studied, Previous_scores, Exam_score) were transformed by discretizing their continuous values into categorical ranges. This operation was performed to group observations into meaningful intervals, facilitating the visualization of patterns within specific value ranges.

- **Hours_studied** varies from 1 to 44 and was divided in 4 groups, from 1-10,11-20,21-30,31-44 and each group was respectively associated the numbers 1 through 4.
- **Previous_scores** varies from 50 to 100 and to create 3 different classes we analyzed the quantiles to understand which thresholds use. With the first quantile being 63 and third quantile being 88, the students with a grade in between the quantiles were assigned the value 2, the ones below the first 1 and the ones with a grade higher than 88, 3.
- **Exam_score** varies from 55 to 101 and, as it's a different exam, it's necessary to re-evaluate and change the thresholds of the different groups. Compared to the previous scores this grades were generally lower; in fact the third quantile drops down to 80 while the first slightly changes to 65. The values assigned are now 1 to grades lower than 65, 2 to grades lower than 80 and 3 for the rest.

Study hours positive impact on students performances

This scatter plot shows every exam scores grouped by how many hours they studied and colored by the student motivational level.

- We can see that usually, as expected, the more hours students spend studying the higher the score tends to be.
- However the variable Motivation level doesn't really add up, we expect to see more green as we move to the right but it's equally distributed in every column.
- This visualization certainly suggest that there is a positive correlation between hours and score but still suggest that there could be other factors affecting students' results.



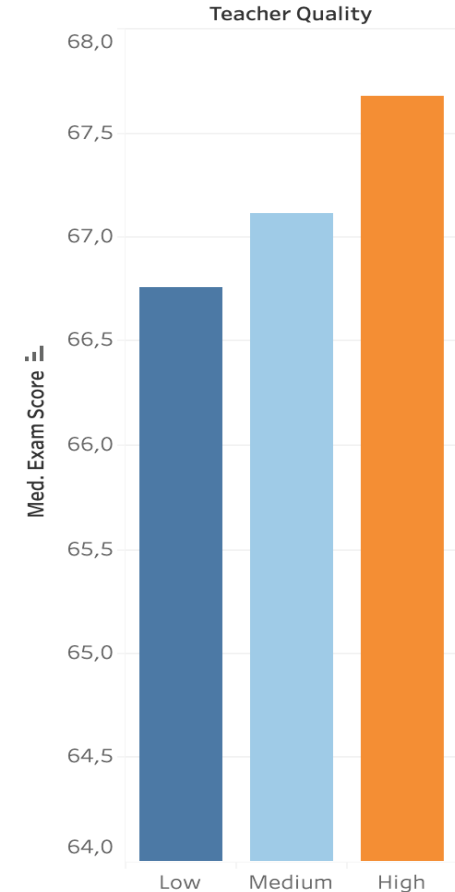
Perceived Teacher Quality effects positively Exam Scores

The chart shows a clear increasing trend in the average exam score across the three levels of perceived teacher quality.

- Students who report a high level of teacher quality achieve the highest average score, followed by those reporting medium quality, while the lowest average performance is observed among students perceiving low teacher quality.
- This pattern suggests a positive association between perceived teaching quality and student performance: higher perceived quality of instruction is associated with higher average exam scores.
- Even if the distance between the different averages isn't that big, it being calculated with more than 6000 records really shows that the teachers' role is still key in having good results.
- In light of what was discussed earlier if the teacher leaves a good impression of themselves and of the subject to the students, they will be encouraged to spend more time studying, which eventually tends to lead to better results.

Bar chart made with Tableau

Teacher quality perceptions and grades

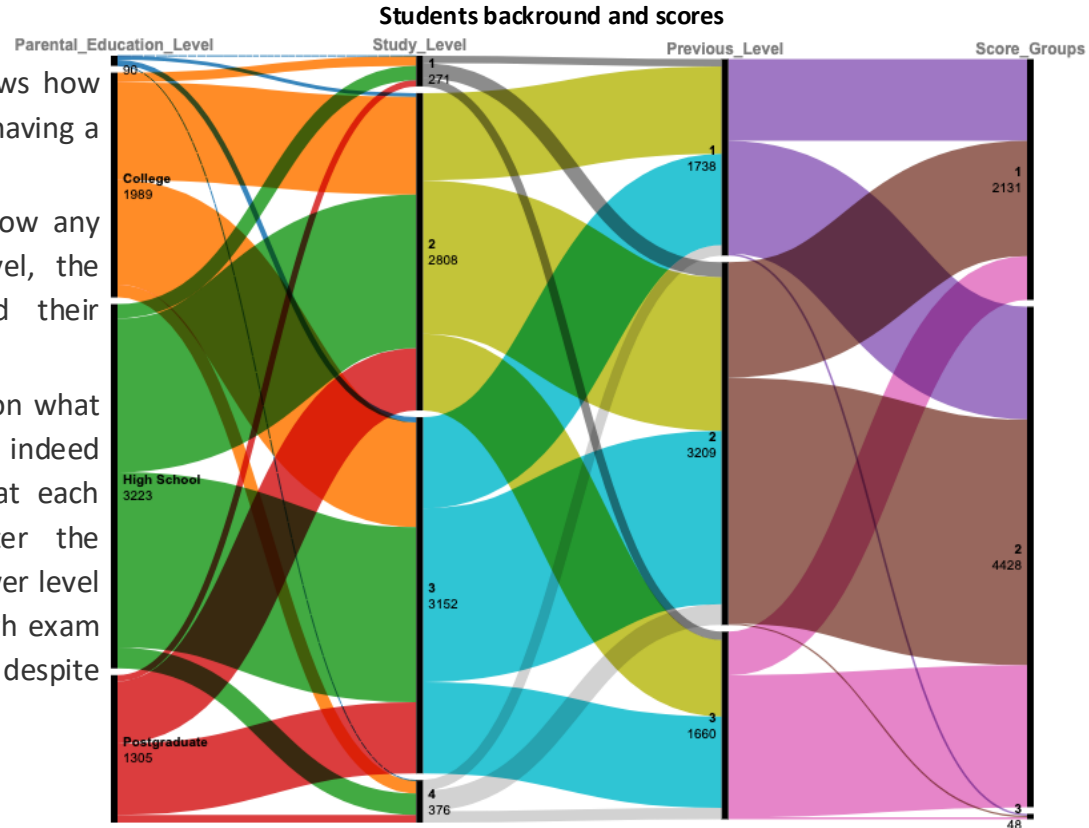


Students path towards great results

This alluvial diagram has 4 different variables and shows how many records for each value of each variable ends up having a certain value in the next variables.

This graph is really confused and doesn't seem to show any clear pattern between the parents' educational level, the amount of hours studied by their children and their performances.

Given this fact it often happens that in data visualization what isn't shown in the graph is just as important as what is; indeed the fact that there isn't any real pattern means that each student is different from every other, no matter the background. In fact, students whose parents have a lower level of education may still study extensively and achieve high exam scores, while other students may obtain poor results despite having spent many hours studying.



LICENCE

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